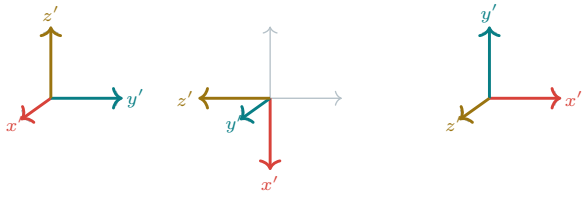
ALGEBRA LADDER 8 OF 9

Rotations in the plane and in space

Turns in the plane commute, turns in space do not, and quaternions keep track of the difference.

1 / THE RULE AND THE PICTURE

Turn a quarter about the x-axis and then a quarter about the y-axis, or the other way round



start x then y y then x

In the plane a turn by θ followed by a turn by φ is a turn by $\theta+\varphi$ in either order, but in space

$$R_y(90^\circ)R_x(90^\circ) \neq R_x(90^\circ)R_y(90^\circ)$$

and the pictures show where the three axes end up in each order.

2 / THE MATRICES

Quarter turns in space, and their two products

$$R_x=\begin{pmatrix}1&0&0\\0&0&-1\\0&1&0\end{pmatrix},\quad R_y=\begin{pmatrix}0&0&1\\0&1&0\\-1&0&0\end{pmatrix},\quad R_yR_x=\begin{pmatrix}0&1&0\\0&0&-1\\-1&0&0\end{pmatrix},\quad R_xR_y=\begin{pmatrix}0&0&1\\1&0&0\\0&1&0\end{pmatrix}.$$

In the plane, $R(30^\circ)R(60^\circ)=R(60^\circ)R(30^\circ)=R(90^\circ)$ for every order, because the angles simply add, and this is the reason complex numbers can be multiplied in either order: multiplying by $\cos\theta+i\sin\theta$ is a turn in the plane.

3 / ROTATION PLUS TRANSLATION

Turning about the origin and sliding do not commute either

Rotate the point $(1,0)$ by 90° about the origin and then slide it by $(2,0)$ and it lands at $(2,1)$, while sliding first to $(3,0)$ and then rotating sends it to $(0,3)$.

Moves that combine turns with slides, the motions of a rigid body, therefore form another non-commuting system, and a computer handles them as a single larger matrix.

4 / QUATERNIONS

Three square roots of -1 that do not commute

Hamilton's quaternions have $i^2=j^2=k^2=ijk=-1$, so that $ij=k$ and $ji=-k$. A turn of 90° about a unit axis is the quaternion $\frac{1}{\sqrt{2}}(1+\text{axis})$, and a vector v is turned to qvq^{-1}, which for $q=\frac{1}{\sqrt{2}}(1+k)$ sends i to j.

The two quarter turns above become $q_yq_x=\frac{1}{2}(1+i+j-k)$ and $q_xq_y=\frac{1}{2}(1+i+j+k)$, which differ only in the sign of k and are the two rotations of 120° about the axes $(1,1,-1)$ and $(1,1,1)$.

5 / TRY THESE

Compare the orders

(a) Check $ij=k$ and $ji=-k$ with the rule $(a+bi)(c+di)$ extended by $ij=k$; swapping the order flips the sign of k.

(b) Hold a book flat and give it a quarter turn about the vertical and then about the long edge, then reverse the order, and compare where the cover points.

(c) Check that two turns about the same axis commute: $R_x(90^\circ)R_x(180^\circ)=R_x(180^\circ)R_x(90^\circ)=R_x(270^\circ)$.

ASIDE / WHY THREE DIMENSIONS CHANGE THE RULES

The only ways to multiply arrays of real numbers with division are the reals, the complex numbers and the quaternions, and commutativity is lost at the quaternions (Frobenius).

LINKS TO OTHER SHEETS

The previous sheet shows two dimensional matrices that fail to commute, and the last sheet collects every structure met in the ladder and the laws each one keeps.

Convention. R_x, R_y turn about the axes; R_yR_x means R_x acts first; q unit quaternion; x', y', z' are the images of the axes after the turns.